

QVerse — Week 13: Hybrid Quantum–Classical Workflows

Theory Booklet | Introduction to Quantum Computing & Programming

QVerse

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How this booklet fits the course

This theory booklet is designed to be read alongside `13_hybrid_variational_workflows.ipynb`. It contains the conceptual and mathematical material that should be explained during Week 13; the notebook then turns those ideas into experiments, circuit construction, and exercises.

Learning objectives

- Build parameterized quantum circuits.
- Compute expectation values with `StatevectorEstimator`.
- Plot a cost landscape.
- Implement a simple classical optimization loop.

Recommended workflow

1. Read the relevant theory section.
2. Predict the result of the corresponding notebook experiment.
3. Run the notebook.
4. Explain any difference between the prediction and the output.
5. Complete the core exercises without treating the solution notebook as a substitute for reasoning.

1. Why hybrid quantum-classical methods exist

Current quantum processors are limited in qubit count, fidelity, connectivity, and circuit depth. Many proposed near-term methods therefore use a **hybrid loop**:

classical parameters $\rightarrow$ quantum circuit $\rightarrow$ measured objective $\rightarrow$ classical update.

The quantum computer evaluates a parameterized state or expectation value. A classical optimizer decides how to change the parameters.

This pattern underlies algorithms such as VQE and QAOA.

2. Parameterized quantum circuits

A parameterized circuit contains symbolic variables such as θ .

A simple ansatz is

$$|\psi(\theta)\rangle = R_y(\theta)|0\rangle.$$

Using Week 7,

$$|\psi(\theta)\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + \sin\left(\frac{\theta}{2}\right)|1\rangle.$$

The word **ansatz** means a chosen family of candidate states.

The optimizer searches this family instead of the entire Hilbert space.

3. Observables and expectation values

An observable A is represented by a Hermitian operator.

For state $|\psi\rangle$,

$$\langle A \rangle = \langle \psi | A | \psi \rangle.$$

For Pauli Z ,

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Therefore

$$\langle Z \rangle = P(0) - P(1).$$

For the R_y ansatz,

$$P(0) = \cos^2 \frac{\theta}{2},$$

$$P(1) = \sin^2 \frac{\theta}{2},$$

so

$$\langle Z \rangle = \cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2} = \cos \theta.$$

This predicts the entire cost landscape before any code runs.

4. Sampler versus estimator thinking

A **sampler** returns samples or a probability distribution over bit strings.

An **estimator** targets expectation values of observables.

For a variational algorithm, the quantity we care about is often an energy or cost function such as

$$E(\theta) = \langle \psi(\theta) | H | \psi(\theta) \rangle.$$

An estimator interface is therefore a natural abstraction.

The notebook uses an ideal statevector estimator so students can focus on the workflow before adding shot noise.

5. Cost functions

A cost function converts circuit behavior into a scalar quantity that a classical algorithm can optimize.

Examples:

- expectation value of Z ;
- energy of a Hamiltonian;
- probability of a target bit string;
- objective function for a combinatorial problem.

A well-defined optimization problem requires:

1. parameters;
2. an ansatz;
3. an observable or measurement rule;
4. a scalar cost;
5. an optimization method.

6. Grid search

For one parameter, a grid search is conceptually transparent:

1. choose values $\theta_1, \theta_2, \dots$;
2. evaluate $E(\theta_j)$;
3. select the smallest value.

It is inefficient in high dimensions, but excellent for learning because the full landscape can be plotted.

For

$$E(\theta) = \langle Z \rangle = \cos \theta,$$

the minimum is

$$E_{\min} = -1$$

at

$$\theta = \pi$$

modulo 2π .

7. A one-qubit Hamiltonian

The notebook uses

$$H = 0.7Z + 0.3X.$$

For the $R_y(\theta)|0\rangle$ ansatz,

$$\langle Z \rangle = \cos \theta,$$

and

$$\langle X \rangle = \sin \theta.$$

Therefore the energy is

$$E(\theta) = 0.7 \cos \theta + 0.3 \sin \theta.$$

This can be rewritten as

$$E(\theta) = R \cos(\theta - \delta),$$

where

$$R = \sqrt{0.7^2 + 0.3^2} = \sqrt{0.58}.$$

So the exact minimum is

$$E_{\min} = -\sqrt{0.58} \approx -0.7616.$$

This analytic result provides a useful benchmark for the numerical grid search.

8. Hamiltonians as sums of Pauli terms

Many quantum Hamiltonians are written as sums of Pauli strings:

$$H = \sum_j c_j P_j,$$

where each P_j is a product of I, X, Y, Z operators across qubits.

Then

$$\langle H \rangle = \sum_j c_j \langle P_j \rangle.$$

This decomposition is central to VQE because each Pauli term can be estimated from measurements in an appropriate basis.

The one-qubit $0.7Z + 0.3X$ example is the simplest version of this idea.

9. VQE concept

The Variational Quantum Eigensolver seeks an approximate ground-state energy of a Hamiltonian.

The variational principle states that for any normalized trial state,

$$\langle \psi(\theta) | H | \psi(\theta) \rangle \geq E_0,$$

where E_0 is the true ground-state energy.

Therefore minimizing the expectation value over an ansatz family gives an upper bound to the ground-state energy.

A practical VQE loop is:

1. prepare $|\psi(\theta)\rangle$;
2. estimate Hamiltonian terms;
3. compute energy;
4. classical optimizer proposes new θ ;
5. repeat.

10. QAOA concept

QAOA is another hybrid variational framework, aimed mainly at combinatorial optimization.

It alternates parameterized evolutions under:

- a cost Hamiltonian;
- a mixing Hamiltonian.

The parameters are optimized classically.

The Week 13 notebook does not implement full QAOA, but students should recognize the same hybrid architecture.

11. Why the ansatz matters

An ansatz that is too simple may not be able to represent the target state.

An ansatz that is too deep may be difficult to optimize and too noisy for hardware.

Relevant properties include:

- expressibility;
- entangling structure;
- parameter count;
- circuit depth;

- compatibility with hardware connectivity;
- optimization landscape.

Choosing an ansatz is therefore an engineering decision, not only a mathematical one.

12. Optimization challenges

Real variational algorithms can face:

- local minima;
- flat regions;
- noisy objective values;
- finite-shot fluctuations;
- parameter correlations;
- hardware drift;
- barren plateaus in some large ansatz families.

The notebook deliberately uses a one-parameter landscape so students can see the full optimization geometry before encountering these complications.

13. Gradients

Many optimizers can use derivatives

$$\frac{\partial E}{\partial \theta}.$$

For certain quantum gates, gradients can be estimated using a **parameter-shift rule**, for example

$$\frac{\partial E}{\partial \theta} = \frac{1}{2} \left[E\left(\theta + \frac{\pi}{2}\right) - E\left(\theta - \frac{\pi}{2}\right) \right]$$

for appropriate gate generators and cost functions.

This is not required in the notebook but is a useful bridge to more advanced variational work.

14. Common misconceptions

“Variational means the quantum computer performs the optimization”

Usually a classical optimizer updates the parameters.

“A lower cost always proves a quantum advantage”

No. It only shows success on the chosen objective. Classical baselines are still needed.

“More expressive ansatz is always better”

Greater expressibility can increase depth and make optimization harder.

“Estimator and sampler are interchangeable concepts”

They answer different questions: expectation values versus sampled bit strings.

15. What to know before the notebook

You should be able to write the hybrid loop, derive

$$\langle Z \rangle = \cos \theta$$

for $R_y(\theta)|0\rangle$, and explain why

$$E(\theta) = \langle \psi(\theta) | H | \psi(\theta) \rangle$$

can be minimized by a classical optimizer.

You should also understand the roles of ansatz, observable, cost function, and baseline.

Suggested references

- IBM Quantum Learning, *Basics of Quantum Information* and current Qiskit learning material: <https://quantum.cloud.ibm.com/learning/en/courses>
- IBM Quantum Documentation and Qiskit guides: <https://quantum.cloud.ibm.com/docs>
- Quantum Rings, *Quantum 101*: <https://www.quantumrings.com/course/quantum101>
- M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press.

The software-facing details in the practical notebook should be treated as the canonical implementation for this course. The purpose of this booklet is to explain the physics, mathematics, and algorithmic reasoning behind that implementation.